



Week 6 Lecture 4

▼ Class	BSCCS2003
🕒 Created	@October 11, 2021 8:07 PM
🔗 Materials	
# Module #	35
▼ Type	Lecture
☰ Week #	6

REST APIs

Example #2: CoWin public APIs

- For Co-Win app: Vaccine registration and information
- Unauthenticated APIs:
 - State-wise search, districts, etc.
- Authenticated APIs:
 - Book appointment

<https://apisetu.gov.in/public/marketplace/api/cowin#/>

General Information

Servers

https://cdn-api.co-vin.in/api - Production Server

Authorize

User Authentication APIs

Metadata APIs

Appointment Availability APIs

Certificate APIs

Schemas

CentersSchema

Example: Availability API

GET

/v2/appointment/sessions/public/findByPin

Get vaccination sessions by PIN

API to get planned vaccination sessions on a specific date in a given pin.

Parameters

Try it out

Name	Description
Accept-Language string (header)	The locate code of the preferred language such as en_US. The text data will be returned in the preferred language along with default English text. Example : hi_IN hi_IN
pincode * required string (query)	110001
date * required string (query)	31-03-2021

Testing public API

WARNING: Do NOT overdo this (This is a public API related to health services, overloading this might have some serious implications especially on someone's life)

```
$ curl -X GET "https://cdn-api.co-vin.in/api/v2/appointment/sessions/public/findByPin"
-H "accept: application/json" -H "Accept-Language: en_US"

{
  "errorCode": "USRRES0001",
  "error": "Input parameter missing"
}
```

```
$ curl -X GET "https://cdn-api.co-vin.in/api/v2/appointment/sessions/public/findByPin?pincode=600020&date=04-08-2021"
-H "accept: application/json" -H "Accept-Language: en_US"
```

```
{"sessions":[
{"center_id":604384,
"name":"Fortis Malar Hospital",
"address":"Fortis Malar HospitalChennai TN.",
"state_name":"Tamil Nadu",
"district_name":"Chennai",
"block_name":"Adyar",
"pincode":600020,
"from":"13:30:00", "to":"15:30:00", "lat":12, "long":80, "fee_type":"Paid",
"session_id":"d40bd2c9-0f42-4948-b794-e3c31fa7c3cc",
"date":"04-08-2021",
"available_capacity":85,
"available_capacity_dose1":40,
"available_capacity_dose2":45,
"fee":"1250", . . .
```

```
curl -X GET
"https://cdn-api.co-vin.in/api/v2/registration/certificate/public/download?ben
eficiary_reference_id=1234567890123" -H "accept: application/json" -H
"Accept-Language: en_US" -H "User-Agent: Mozilla/5.0"
```

Unauthenticated access!

Authentication

- Many APIs must be protected:
 - Only meant for specific users
 - Avoid abuse by overloading servers

How to implement this?

Require a "token" that only a valid user can have

- Securely give token only when the user logs in - Google OAuth, Facebook, etc.
- **API key:** One time token that user downloads - can be copied, so potentially less secure unless combined with other methods

Summary

- **API examples:** CoWin, Google Cloud, Twitter, GitHub,... etc.
- Authentication may be enforced or optional on some parts
- Allows 3rd party integrations
- Equivalent of a "remote procedure call" — call a function on a remote system